



SAINT IGNATIUS' COLLEGE

Trial Higher School Certificate

2002

MATHEMATICS

EXTENSION 1

2:00am – 4:05 pm
Friday 30th August 2002

General Instructions

- Reading time : 5 minutes
- Working time: 2 hours
- Write using blue or black pen
- *Write your name and teacher's name on each answer booklet*
- Board approved calculators may be used
- A table of standard integrals is provided
- Total Marks (84)
- Attempt Questions 1 – 7
- All questions are of equal value

Students are reminded that this is a trial examination only and cannot in any way guarantee the content or the format of the 2002 Mathematics Extension 1 Higher School Certificate examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, x > 0$

Total marks (84)

Attempt Questions 1 – 7

All questions are of equal value

Answer each question in a SEPARATE writing booklet.

QUESTION 1 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Factorise $64 - y^3$. 1
- (b) Find the coordinates of the point that divides the join of $A(1, -3)$ and $B(-5, 7)$ externally in the ratio $5 : 2$. 2
- (c) Solve $\frac{2x}{x-2} > 1$. 3
- (d) Find the acute angle, to the nearest degree, between the lines $3x - y + 2 = 0$ and $x + 2y - 5 = 0$. 3
- (e) Using the expansion of $\cos(A + B)$, find the exact value of $\cos 75^\circ$ in surd form with a rational denominator. 3

QUESTION 2 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Find the general solution of the equation $\sin^2 x - 2 \cos x + 2 = 0$, where x is in radians. 3
- (b) Find $\int \cos^2 2x \, dx$. 2
- (c) Use the substitution $x = u^2 - 1$ for $u > 0$, to evaluate $\int_3^8 \frac{x-1}{\sqrt{x+1}} \, dx$. 4
- (d) The probability that a student will attend university after doing the HSC is 0.7. In a class of 25 students, what is the probability that at least 23 students will proceed to university after the HSC? 3

QUESTION 3 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Find the indefinite integral $\int \frac{dx}{9+16x^2}$. 2
- (b) Consider the function $f(x) = \cos^{-1}(x+2) - \frac{\pi}{2}$.
- (i) What is the domain of $y = f(x)$? 1
- (ii) Sketch the graph of $y = f(x)$. 2
- (c) Consider the function $f(x) = \tan^{-1} x + \tan^{-1} \frac{1}{x}$.
- (i) Prove the derivative of $f(x)$ is zero. 2
- (ii) What is the domain of $y = f(x)$? 1
- (iii) Find the value of $f(x)$ over the domain of $y = f(x)$. 2
- (d) The velocity v of a particle moving in a straight line at position x is given by: 2

$$v = 1 + 2x.$$

Find the acceleration of the particle at position x .

QUESTION 4 (12 marks) Use a SEPARATE writing booklet.

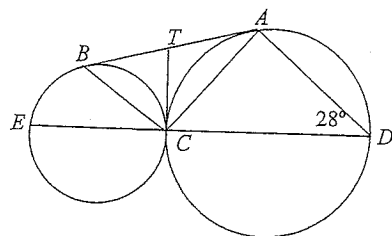
Marks

- (a) Consider the letters of the word MILLER.
- (i) How many arrangements of these letters are possible if the letters are arranged in a straight line? 1
- (ii) What is the probability that the L's will be separated when the letters are arranged in a straight line? 2
- (iii) If the letters are arranged in a circle, how many arrangements are possible? 1
- (iv) If the letters are arranged in a circle, what is the probability the L's will be opposite each other? 2
- (b) Find the term independent of x in the expansion of $\left(x - \frac{2}{x^2}\right)^9$, expressing your answer as an integer. 3
- (c) Prove that $\binom{n+1}{2} + \binom{n+2}{2}$ is the square of an integer. 3

QUESTION 6 (12 marks) Use a SEPARATE writing booklet.

Marks

(a)

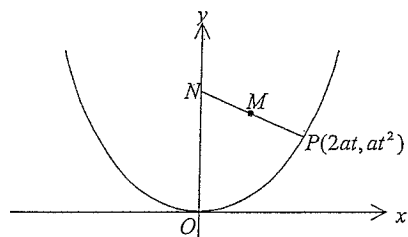


4

AB is a common tangent to the circles, which touch at C . CT is the tangent at C . A line through C meets the circles at D and E . $\angle ADC = 28^\circ$.

Find the size of $\angle ACB$, giving reasons.

(b)



A normal is drawn to the parabola $x^2 = 4ay$ at a variable point $P(2at, at^2)$ on the parabola. The normal meets the y axis at N . M is the midpoint of PN . The coordinates of M are $(at, at^2 + a)$ and O is the origin.

(i) Show that the equation of the locus of M is the parabola $ay = x^2 + a^2$. 1

(ii) What is the vertex of the parabola $ay = x^2 + a^2$? 1

(iii) What is the domain of this locus? Explain. 1

(c) The rate of change of the amount of pollutant (x units) in a lake after t days of rain is given by: $\frac{dx}{dt} = \frac{1}{4} - \frac{x}{16}$.

(i) Show that $x = 4 + Ae^{-\frac{t}{16}}$ satisfies this equation. 2

(ii) If the initial amount of pollutant is 20 units, find the value of A . 1

(iii) In how many days will the amount of the pollutant drop to 8 units? 2

QUESTION 5 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Two of the roots of the equation $x^3 + 6x^2 - x + k = 0$ are 2, -3, where k is a constant.

(i) By considering the sum of the roots of this equation, find the third root. 2

(ii) Hence or otherwise find the value of the constant k . 2

(b) Use one step of Newton's method to find an approximation to the root of the equation $x^4 - 10x + 7 = 0$ near $x = 2$. 3

(c) A particle moves in Simple Harmonic Motion about an origin O . The period of motion is 4 seconds and the amplitude is 6 cm.

(i) Write down an expression for x in terms of t , when the particle is x cm from O , t seconds after passing through O . 2

(ii) Find its speed as it passes through O . 2

(iii) What is the maximum acceleration of the particle? 1



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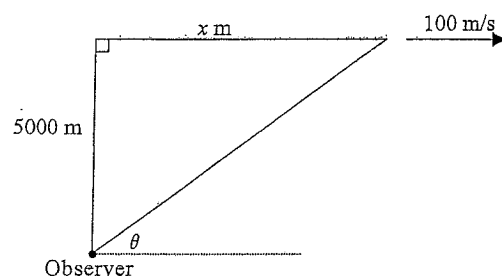
Suggested Solutions

QUESTION 7 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Prove by mathematical induction, where n is a positive integer and $n \geq 2$, 5
that $\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$.

(b)



At a certain instant, a plane flies overhead at a constant altitude of 5000 metres, at a constant speed of 100 metres per second.
When the plane has travelled x metres from the overhead position, its angle of elevation from the observer is θ radians.

- (i) Show that $\frac{dx}{d\theta} = -\frac{5000}{\sin^2 \theta}$. 2
- (ii) Hence show that $\frac{d\theta}{dt} = -\frac{1}{50} \sin^2 \theta$. 2
- (iii) Find the rate at which the angle of elevation is changing:
- (α) when the plane is overhead. 1
- (β) 50 seconds after the plane is overhead. 2

End of Examination

Mathematics Extension 1: Question 1		
Suggested Solutions	Marks Awarded	Marker's Comments
(a) $64 - y^3 = (4 - y)(16 + 4y + y^2)$ (1)		
(b) $A(1, -3) \quad B(-5, 7) \quad S: -2$ $\left(\frac{5 \times (-5) + (-2) \times 1}{5 + (-2)}, \frac{5 \times 7 + (-2) \times (-3)}{5 + (-2)} \right)$ $(-9, 13\frac{2}{3})$ (2)		
(c) $\frac{2x}{x-2} > 1$ $2x(x-2) > (x-2)^2$ $2x^2 - 4x > x^2 - 4x + 4$ $x^2 > 4$ $x < -2, x > 2$ (3)		
(d) $3x - y + 2 = 0: m = 3$ $x + 2y - 5 = 0: m = -\frac{1}{2}$ $\tan \theta = \left \frac{3 - (-\frac{1}{2})}{1 + 3 \times (-\frac{1}{2})} \right $ $= 7$ $\theta = 82^\circ$ (nearest degree) (3)		
(e) $\cos(A+B) = \cos A \cos B - \sin A \sin B$ $\cos 75^\circ = \cos(45^\circ + 30^\circ)$ $= \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ$ $= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2}$ $= \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$ $= \frac{\sqrt{6}-\sqrt{2}}{4}$ (1.5)		

Mathematics Extension 1: Question 2		
Suggested Solutions	Marks Awarded	Marker's Comments
(a) $\sin^2 x - 2 \cos x + 2 = 0$ $1 - \cos^2 x - 2 \cos x + 2 = 0$ $\cos^2 x + 2 \cos x - 3 = 0$ $(\cos x + 3)(\cos x - 1) = 0$ $\cos x = -3 : \text{no solution}$ $\cos x = 1 : x = 2n\pi$ where n is an integer. (3)		
(b) $\int \cos^2 2x \, dx = \int \frac{1}{2} (1 + \cos 4x) \, dx$ $= \frac{1}{2} (x + \frac{1}{4} \sin 4x) + C$ (2)		
(c) $\int_3^8 \frac{x-1}{\sqrt{x+1}} \, dx$ $= \int_2^3 \frac{u^2-2}{u} \times 2u \, du$ $= \int_2^3 (2u^2-4) \, du$ $= \left[\frac{2}{3} u^3 - 4u \right]_2^3$ $= (18-12) - (5\frac{1}{3}-8)$ $= 8\frac{2}{3}$ $x = u^2 - 1$ $dx = 2u \, du$ $x = 3 \Rightarrow u = 2$ $x = 8 \Rightarrow u = 3$ (4)		
(d) $P(\text{at least 23 students})$ $= P(23) + P(24) + P(25)$ $= \binom{25}{23} (0.7)^{23} (0.3)^2 + \binom{25}{24} (0.7)^{24} (0.3) + (0.7)^{25}$ $= 0.00896$ (3)		

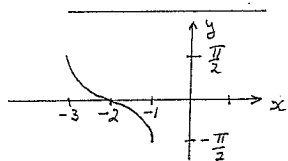
Mathematics Extension 1: Question 3

Suggested Solutions

$$\begin{aligned} (a) \int \frac{dx}{9+16x^2} &= \int \frac{dx}{16\left(\frac{9}{16}+x^2\right)} \\ &= \frac{1}{16} \times \frac{1}{\frac{3}{4}} \tan^{-1} \frac{2x}{\frac{3}{4}} + C \\ &= \frac{1}{12} \tan^{-1} \frac{4x}{3} + C \end{aligned}$$

OR Use substitution $u = 4x$.

$$\begin{aligned} (b) \quad f(x) &= \cos^{-1}(x+2) - 1 \\ (i) \text{ Domain: } &-1 \leq x+2 \leq 1 \\ &\text{i.e. } -3 \leq x \leq -1 \end{aligned}$$



$$\begin{aligned} (c) \quad f(x) &= \tan^{-1}x + \tan^{-1}\frac{1}{x} \\ (i) \quad f'(x) &= \frac{1}{1+x^2} + \frac{1}{1+(\frac{1}{x})^2} \times \left(-\frac{1}{x^2}\right) \\ &= \frac{1}{1+x^2} - \frac{1}{x^2+1} \\ &= 0 \end{aligned}$$

(ii) Domain: all x , except $x=0$

$$\begin{aligned} (iii) \text{ For } x > 0: f(1) &= \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2} \\ \text{For } x < 0: f(-1) &= -\frac{\pi}{4} + \left(-\frac{\pi}{4}\right) = -\frac{\pi}{2} \\ \therefore f(x) &= \frac{\pi}{2} \text{ for } x > 0; f(x) = -\frac{\pi}{2} \text{ for } x < 0. \end{aligned}$$

$$\begin{aligned} (d) \quad v &= 1+2x \\ a &= \frac{dv}{dx} \left(\frac{1}{2}v^2\right) \\ &= \frac{dv}{dx} \cdot \frac{1}{2} (1+2x)^2 \\ &= \frac{1}{2} \times 2(1+2x) \times 2 \\ &= 2(1+2x). \end{aligned}$$

Mathematics Extension 1: Question 4

Suggested Solutions

Marks
Awarded

Marker's Comments

(a) MILLER.

$$(i) \text{ No. of arrangements} = \frac{6!}{2!} = 360 \quad (1)$$

$$\begin{aligned} (ii) \text{ Prob (L's separated)} \\ &= 1 - P(\text{L's together}) \\ &= 1 - \frac{5!}{360} \\ &= \frac{2}{3} \quad (2) \end{aligned}$$

$$(iii) \text{ No. of arrangements} = \frac{5!}{2!} = 60 \quad (1)$$

$$\begin{aligned} (iv) P(\text{L's opposite}) &= \frac{1}{5} \\ \text{or } \frac{4!}{5!} &= \frac{1}{5} \quad (2) \end{aligned}$$

$$(b) \left(x - \frac{2}{x^2}\right)^9.$$

$$\begin{aligned} \text{Term indep. of } x &= \binom{9}{3} x^6 \left(-\frac{2}{x^2}\right)^3 \\ (\text{by inspection}) &= 504 \times (-8) \times x^0 \\ &= -672 \end{aligned}$$

$$\begin{aligned} \text{OR } T_{r+1} &= \binom{9}{r} x^{9-r} \left(-\frac{2}{x^2}\right)^r \\ &= \binom{9}{r} (-2)^r x^{9-3r} \quad (3) \end{aligned}$$

$$\begin{aligned} (c) \binom{n+1}{2} + \binom{n+2}{2} &= \frac{(n+1)n}{2} + \frac{(n+2)(n+1)}{2} \\ &= \frac{n+1}{2} [n+n+2] \\ &= \frac{n+1}{2} \times 2(n+1) \\ &= (n+1)^2 \end{aligned}$$

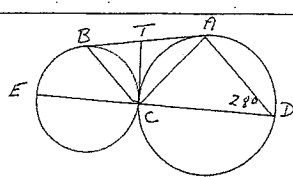
which is the square of an integer.

(3)

Mathematics Extension 1: Question 5

Suggested Solutions	Marks Awarded	Marker's Comments
(a) $x^3 + 6x^2 - x + k = 0$ (i) Let the third root be α. $2 + (-3) + \alpha = -6$ $\alpha = -5$ (2) (ii) By product of roots: $2 \times (-3) \times (-5) = -k$ $k = -30$ (2)		
(b) $f(x) = x^4 - 10x + 7$ $f'(x) = 4x^3 - 10$ $x_1 = 2 - \frac{f(2)}{f'(2)}$ $= 2 - \frac{3}{22}$ $= 1.86$ (2 d.p.) (3)		
(c) Period = 4 sec, Amplitude = 6 cm. (i) $T = \frac{2\pi}{n} = 4 \therefore n = \frac{\pi}{2}$. $x = 6 \sin \frac{\pi}{2} t$. (2) (ii) $v = 6 \times \frac{\pi}{2} \cos \frac{\pi}{2} t$. When $t=0$, $v = 3\pi$. Speed is 3π cm/s at 0. (2) (iii) $a = -3\pi \times \frac{\pi}{2} \sin \frac{\pi}{2} t$ $= -\frac{3\pi^2}{2} \sin \frac{\pi}{2} t$ Maximum acceleration is $\frac{3\pi^2}{2}$ cm/sec² (1)		

Mathematics Extension 1: Question 6

Suggested Solutions	Marks Awarded	Marker's Comments
(a)  $\angle LTC = 28^\circ$ (alternate seg. theorem) $\angle LTC = 28^\circ$ ($TA = TC$: tangents from an external point are equal) $\angle LTC = 56^\circ$ (exterior angle of $\triangle TAC$). But $TB = TC$ (tangents from ext. point) $\therefore \angle LTC = \frac{1}{2}(180^\circ - 56^\circ)$ (angle sum of isos. $\triangle TBC$) $= 62^\circ$ $\therefore \angle LTC = 28^\circ + 62^\circ = 90^\circ$. (4)		
(b)(i) Coord of M. $x = at$, $y = at^2 + a$. $t = \frac{x}{a} \therefore y = a \times \frac{x^2}{a^2} + a$ $ay = x^2 + a^2$ (1) (ii) $x^2 = a(y - a)$ Vertex is $(0, a)$. (1) (iii) Domain - all x except $x=0$, because normal at $(0,0)$ does not cut the y-axis at one point - it is the y-axis (1)		
(c) $\frac{dx}{dt} = \frac{1}{4} - \frac{x}{16}$. (i) $x = 4 + Ae^{-\frac{t}{16}}$ $\frac{dx}{dt} = -\frac{1}{16} Ae^{-\frac{t}{16}} = -\frac{1}{16}(x - 4) = \frac{1}{4} - \frac{x}{16}$. (2) (ii) $t=0, x=20$: $20 = 4 + Ae^0 \therefore A = 16$ (1) (iii) $x = 4 + 16e^{-\frac{t}{16}}$ When $x=8$, $8 = 4 + 16e^{-\frac{t}{16}}$ $e^{-\frac{t}{16}} = \frac{1}{4}$, $e^{\frac{t}{16}} = 4$ $t = 16 \ln 4 = 22.18$ Approx 22 days. (2)		

Mathematics Extension 1: Question 7(a)

Suggested Solutions

Marks
Awarded

Marker's Comments

(a)

Prove $(1 - \frac{1}{2^2})(1 - \frac{1}{3^2}) \dots (1 - \frac{1}{n^2}) = \frac{n+1}{2n}$ for $n \geq 2$.

When $n=2$, LHS = $1 - \frac{1}{2^2} = \frac{3}{4}$.

RHS = $\frac{2+1}{2 \times 2} = \frac{3}{4}$

$\therefore$ it is true for $n=2$.

Assume it is true for $n=k$:

i.e. assume $(1 - \frac{1}{2^2})(1 - \frac{1}{3^2}) \dots (1 - \frac{1}{k^2}) = \frac{k+1}{2k}$.

When $n=k+1$,

LHS = $(1 - \frac{1}{2^2})(1 - \frac{1}{3^2}) \dots (1 - \frac{1}{k^2})(1 - \frac{1}{(k+1)^2})$

= $\frac{k+1}{2k} (1 - \frac{1}{(k+1)^2})$ by assumption

= $\frac{k+1}{2k} \times \frac{(k+1)^2 - 1}{(k+1)^2}$

= $\frac{k^2 + 2k + 1 - 1}{2k(k+1)}$

= $\frac{k^2 + 2k}{2k(k+1)}$

= $\frac{k(k+2)}{2k(k+1)}$

= $\frac{k+2}{2(k+1)}$

= $\frac{n+1}{2n}$ when $n=k+1$.

$\therefore$ if it is true for $n=k$,
then it is true for $n=k+1$.

Since it is true for $n=2$,
it is true for $n=3, n=4, \dots$

i.e. it is true for all positive
integers $n \geq 2$

(5)

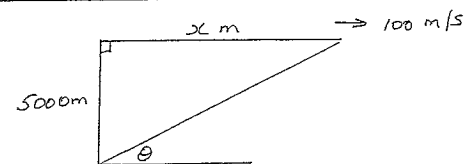
Mathematics Extension 1: Question 7(b)

Suggested Solutions

Marks
Awarded

Marker's Comments

(b)



$$(i) \quad \frac{x}{5000} = \tan(90^\circ - \theta)$$

$$x = 5000 \cot \theta$$

$$\frac{dx}{d\theta} = -5000 \operatorname{cosec}^2 \theta$$

$$= -\frac{5000}{\sin^2 \theta} \quad (2)$$

$$(ii) \quad \frac{dx}{dt} = \frac{dx}{d\theta} \times \frac{d\theta}{dt}$$

$$100 = -\frac{5000}{\sin^2 \theta} \times \frac{d\theta}{dt}$$

$$\frac{d\theta}{dt} = -\frac{100 \sin^2 \theta}{5000}$$

$$= -\frac{1}{50} \sin^2 \theta \quad (2)$$

(iii) (a) When overhead, $\theta = 90^\circ$,

$$\frac{d\theta}{dt} = -\frac{1}{50} \sin^2 90^\circ$$

$$= -\frac{1}{50}$$

Angle is changing at $\frac{1}{50}$ rad/sec. (1)

(b) When $t=50$, $x = 100 \times 50$
 $= 5000$

$$\theta = \frac{\pi}{4}$$

$$\frac{d\theta}{dt} = -\frac{1}{50} \sin^2\left(\frac{\pi}{4}\right)$$

$$= -\frac{1}{100}$$

Angle is changing at $\frac{1}{100}$ rad/sec. (2)